

The Technology Value Stream

Understanding the metrics that measure the efficiency, speed, and reliability of modern software delivery.

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CSD380 DevOps

Module 1.2 Assignment – The Technology Value Stream

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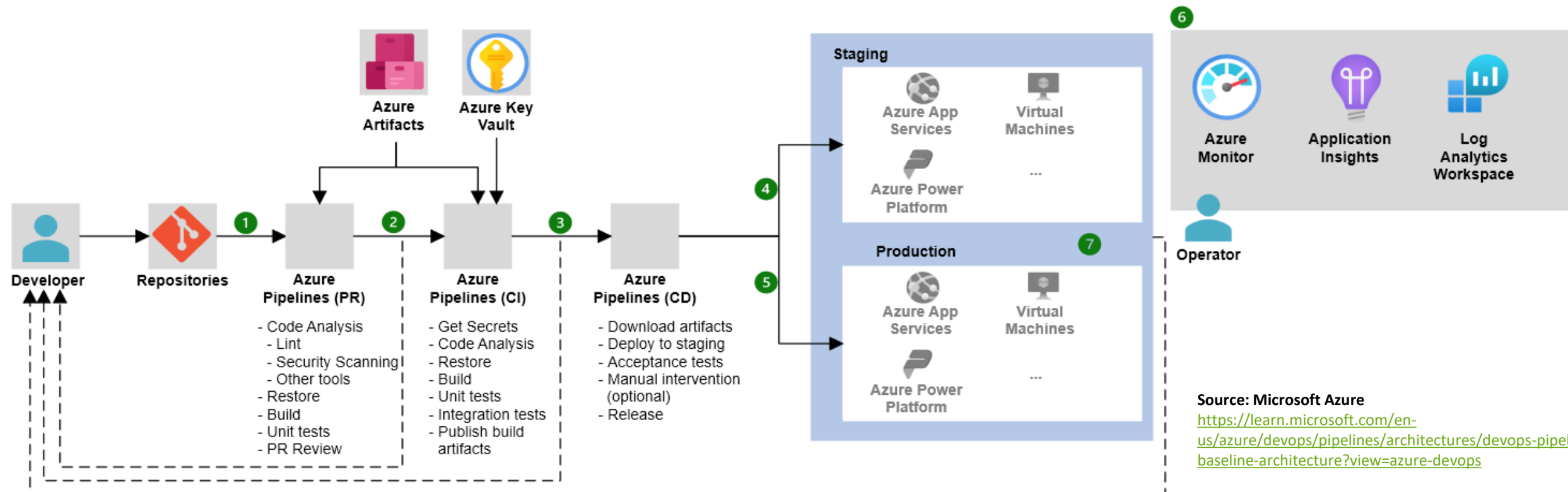
Introduction to the Technology Value Stream

The **Technology Value Stream (TVS)** is the entire sequence of steps that transforms a business hypothesis or customer request into a technology-enabled service that successfully delivers value to the customer.

The TVS encompasses all activities required to deliver software from **concept to production**, including development, testing, deployment, and monitoring phases.

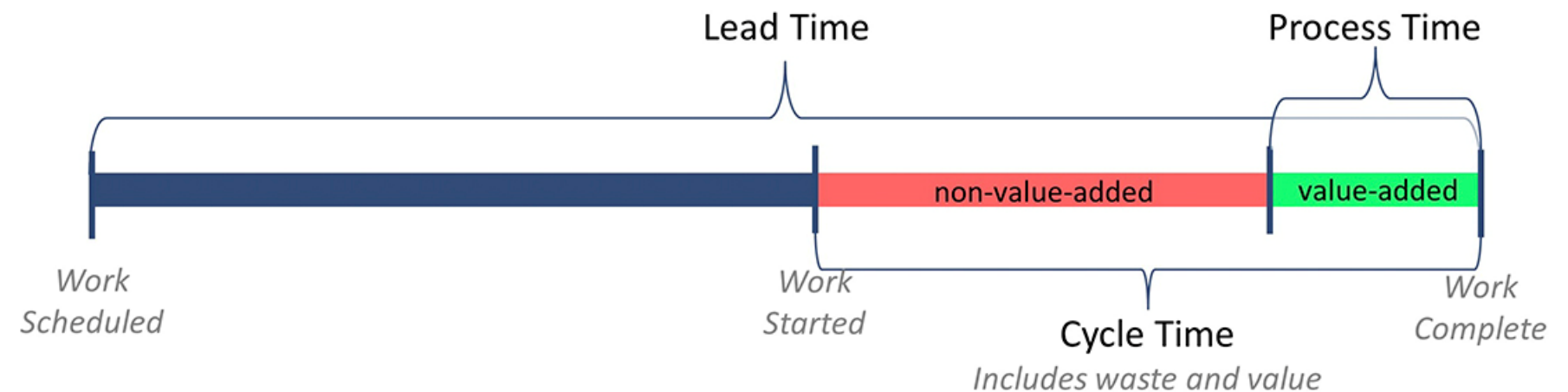
Identifying inefficiencies in the TVS helps teams spot **waste, delays, and bottlenecks** in the software delivery pipeline for improvement.

Key Focus: Optimizing flow, gathering feedback, and enabling continual learning to reduce waste and increase efficiency in the delivery of software value.



Lead Time vs. Processing Time

Understanding the distinction between these core metrics is fundamental to optimizing the TVS. (Source: Page 9 of *The DevOps Handbook, 2nd Edition*)



Key Insight:

- **Lead Time** is the sum of Value Added Time and Non-Value Added Time.
- Reducing the Non-Value Added Time (waiting and handoffs) is the primary lever for shortening Lead Time and delivering value faster.

Metric	Aligned Term	Components
Lead Time (LT)	Total Elapsed Time	The entire duration from when work is requested until it reaches production. This is what your customer experiences.
Processing Time (PT)	Value Added Time	The actual, hands-on time spent actively working on a task—coding, testing, building. This is where value is created.
Queue Time	Non-Value Added Time	Time spent waiting, in handoffs, or due to delays. This is the primary target for improvement.

Source: Kaufman Global
<https://www.kaufmanglobal.com/glossary/lead-time>

Breaking Down Lead Time

Lead time comprises multiple layers. Visual understanding helps teams identify where optimization efforts will have the greatest impact.

In most organizations, queue and waiting time dominate, often representing 90% or more of total lead time. This is where DevOps improvement efforts yield the highest returns.

Processing Time

Active development work

Waiting Time

Delays between activities

Handoff Time

Transitions between teams

Queue Time

Work awaiting approval

The Problem: Deployment Lead Times of Months

1 Sequential handoffs

Work passes through separate teams—requirements, development, QA, ops—with waiting periods between each.

2 Batch-oriented releases

Features accumulate for infrequent, large releases rather than continuous deployment.

3 Manual approval gates

Multiple review steps and change advisory boards create administrative delays.

4 Siloed expertise

Knowledge gaps between teams require additional communication and rework cycles.

The DevOps Ideal

DevOps practices compress deployment lead time to minutes by automating testing, eliminating manual gates, fostering cross-functional collaboration, and implementing continuous deployment pipelines. Teams own their entire delivery cycle from code to production .

(Source: Page 10 of *The DevOps Handbook, 2nd Edition*)



Continuous Integration/Delivery

Seamless integration and frequent, reliable releases.



Workflow Automation

Reduces manual effort and eliminates human error at scale.



Automated Testing

Rapid validation with immediate feedback during development.



Team Collaboration

Improved communication, coordination, and shared ownership.

Result: Rapid feedback loops, frequent releases, improved reliability, and accelerated innovation.

How DevOps Compresses the Value Stream



Modern DevOps practices transform the software delivery pipeline from a slow, sequential process into a fast, continuous flow.

Automate Testing

Catch issues early, reduce manual validation.

1

Deploy Instantly

Infrastructure-as-Code enables on-demand provisioning.

3

Remove Gates

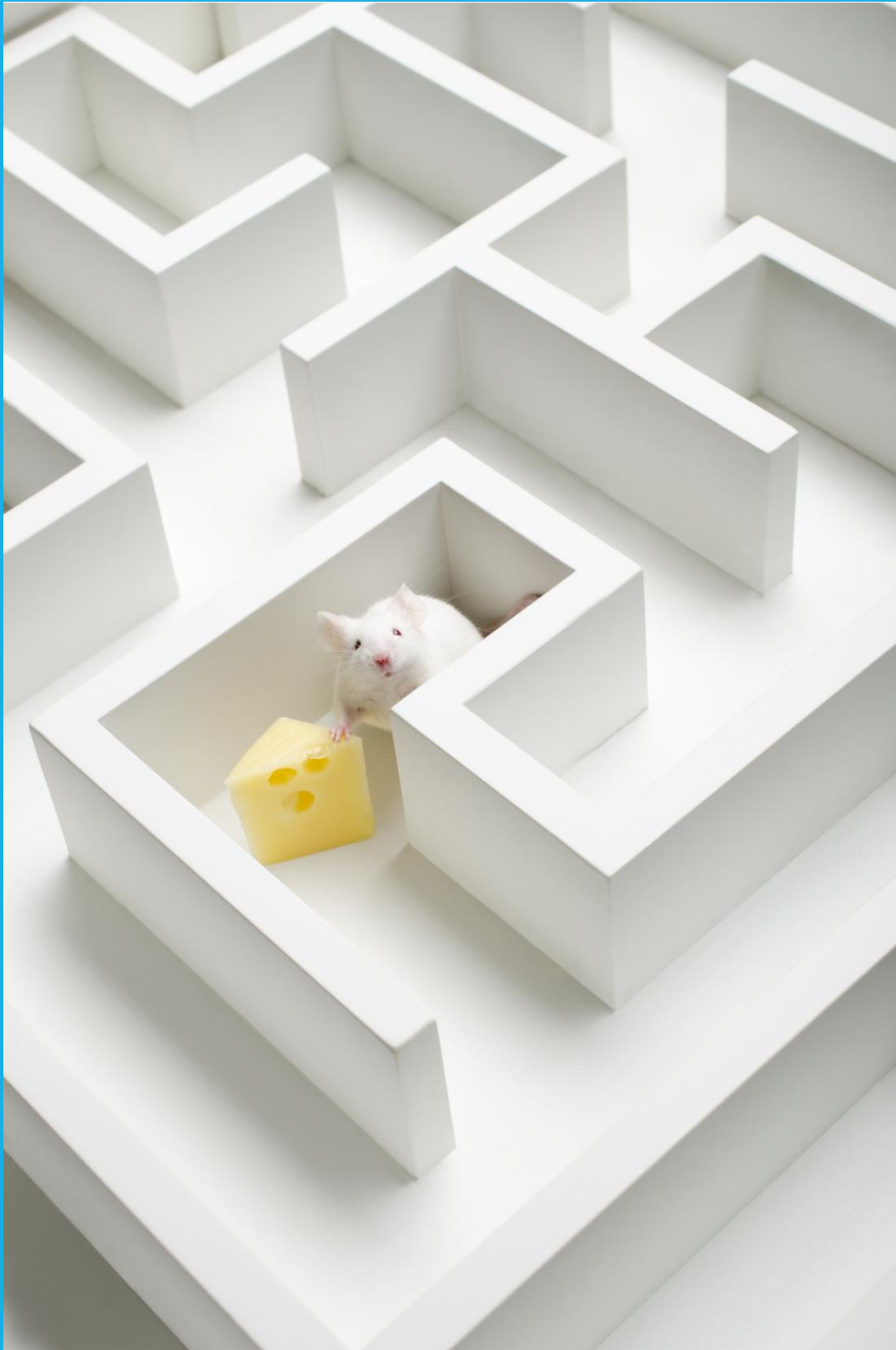
Streamline or automate approvals, eliminate delays.

2

Monitor Continuously

Real-time feedback detects and resolves issues fast.

4



Strategies to Achieve Minute-Level Deployment

Practical, actionable steps to compress your value stream and accelerate software delivery.



Build the CI/CD Pipeline

Automate build, test, and deployment on small, frequent commits.



Shift-Left Testing

Run tests earlier in development, enhance production monitoring and observability.



Break Down Silos

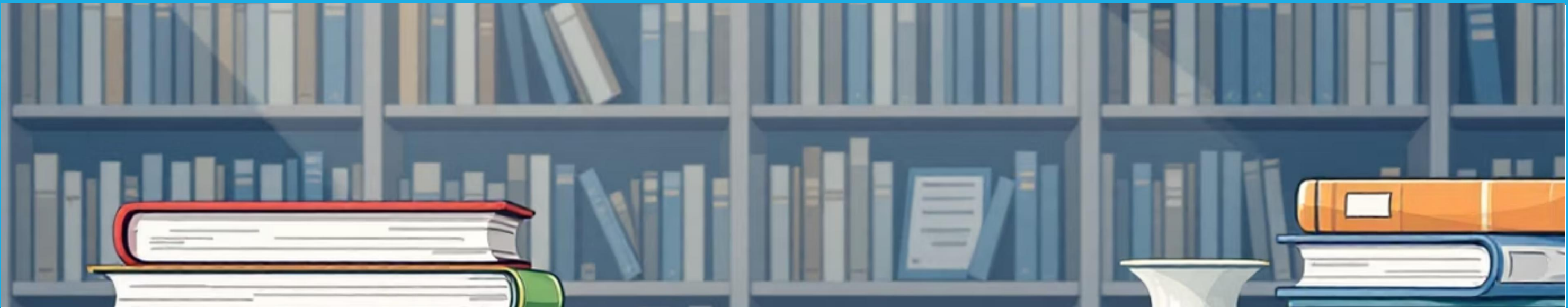
Foster collaboration and shared ownership across development and operations teams.



Measure and Optimize

Track lead time metrics, identify waste, and continuously improve performance.

Source: Kim, G., Humble, J., Debois, P., & Willis, J. (2021). *The DevOps Handbook, 2nd Edition*. O'Reilly Media.



References

Kim, G., Humble, J., Debois, P., & Willis, J. (2021). *The DevOps Handbook, 2nd Edition*. O'Reilly Media.

Microsoft, (n.d.). *CI/CD baseline architecture with Azure Pipelines*. Retrieved October 26, 2025, from <https://learn.microsoft.com/en-us/azure/devops/pipelines/architectures/devops-pipelines-baseline-architecture?view=azure-devops>

Kaufman Global. (n.d.). Lead Time. *Kaufman Global Glossary*. Retrieved October 25, 2025, from <https://www.kaufmanglobal.com/glossary/lead-time/>